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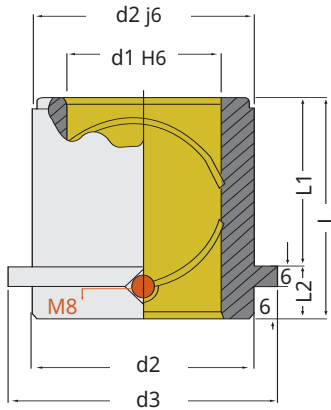
18

Guiding Components

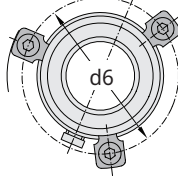


Code: **G36**

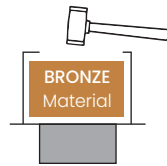
Guide Bush, Bronze Coated & Short Type



G100
Clamping Detail



4 pcs. clamp for
d1 = 38 mm and
larger sizes.



Positioning :
Pls. do not make
direct fitting.
Precision assembly
recommended
with bronze / brass.

Mounting Details:

Code: **G36**

Code: **G35**

d1	d6	d1	d6
20	48.8	20	48.8
25	56.8	25	56.8
30	64.8	30	64.8
40	74.8	40	74.8
50	86.8	50	86.8
63	101.8	63	101.8
80	121.8	80	121.8



Order: **G36**. d1 x L x L1

Material: 1.1213 (Cf53)

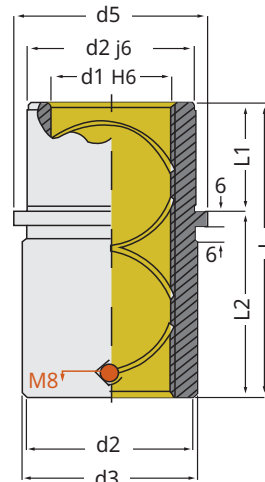
Bronze: SnBz 10 HB 160 - 180

It has helical lubrication and inner lubrication system, its position can be lubricated with grease nipple from outside. Finally, equal distance to all lubrication points is ensured. Due to oil intake feature of bronze coated to inside it, there is no need for frequent lubrication, recommended operating speed is 15-30 mt / min. If some conditions such as lubrication clearance, stroke length, radial load and heat distribution are provided, to reach speed as 600 - 800 stroke per minute is possible.

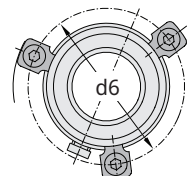


Code: **G35**

Guide Bush, Bronze Coated & Long Type

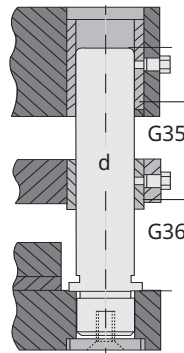


G100
Clamping Detail



4 pcs. clamp for
d1 = 38 mm and
larger sizes.

Mounting Example



IMPORTANT!

Pls. lubricate from
grease nipple
on bush
occasionally.



Order: **G35**. d1 x L x L1

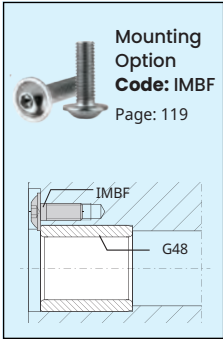
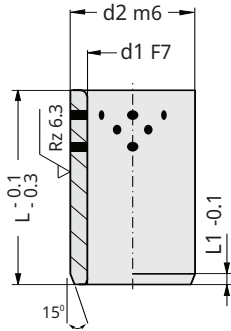
Material: 1.1213 (Cf53)

Bronze: SnBz 10 HB 160 - 180

Hardened steel bushing absorbs strong side thrust. Thus, vibrationless rigid operation is ensured and bronze bush is protected against strong thrust made from end head, inner layer coated bronze with centrifugal technique is resistant against abrasion factor with its high quality. In addition with its excellent heat dissipation feature is ensured rapid distribution of excessive friction temperature.

d1	L	d2	d3	d5	L1	L2
19	43	32	39	40	23	20
20	59					36
24	59	40	46	48	23	36
25	79					56
32	75	48	53	56	30	45
30	93					63
38	82	58	63	66	37	45
40	108					71
48	97	70	77	80	47	50
50	127					80
60	116	85	92	95	60	56
63	150					90
80	120	105	115	118	60	60
	150					90

To prevent reverse closing of the die, 3pcs main dimensions (d1) can be used together with 1pcs intermediate dimension (d1: 19 - 24 - 32 - 38 - 48 - 60 mm).



Guide Bush - Plain / Self-Lubricating

Code: **G48**

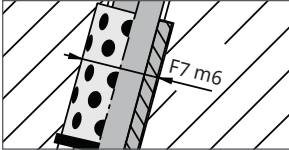
* Hole tolerance for shrink fit: H7

They are self-lubricating bushes, especially suitable for use in straight pillars of die systems and machines such as hydraulic presses and injection moulding machines. They are compatible with areas where magnetic effects are undesirable and have a wide range of applications in industrial use.

Large Stock Availability

d1	d2	L	L1	d1	d2	L	L1	d1	d2	L	L1	d1	d2	L	L1				
8	12	8	2	25	33	20	4	48	60	30	4	150	170	50	4				
		10				25				40				60					
		15				30				50				80					
10	14	10	2			35				40				65		60	70	100	120
		12								45						80		140	
		15								50						160			
12	18	20	2	30	38	20	4	50	60	30	160	180	50	4					
		10				25				40			60						
		12				30				50			80						
		15				35				60			100						
		20				40				70			120						
15	22	12	2	35	40	20	4	60	65	30	4	160	180	50	4				
		16				25				40				60					
		20				30				50				80					
		25				35				60				100					
		30				40				70				120					
16	22	12	2	38	50	25	4	80	96	40	4	160	180	50	4				
		16				30				50				60					
		20				35				60				80					
		25				40				70				100					
		30				50				80				120					
18	28	20	4	40	50	25	4	100	120	50	4	160	180	50	4				
		25				30				40				60					
		30				35				50				70					
		35				40				60				80					
		40				50				70				90					
20	24	20	4	42	55	30	4	120	140	50	4	160	180	50	4				
		25				35				40				50					
		30				40				50				60					
		35				45				60				70					
		40				50				70				80					
24	33	20	4	45	55	30	4	140	160	50	4	160	180	50	4				
		25				35				40				50					
		30				40				50				60					
		35				45				60				70					
		40				50				70				80					

Mounting Example

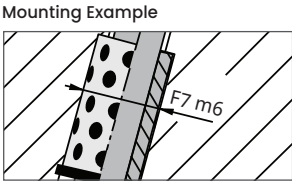


 Order:
G48. d1 x d2 x L

Material:
Bronze + graphite inserts

 Customised
Guide Units

Upon request, Güvenal manufactures highly complex and precise guide elements based on customer specifications and provides measurement protocols if needed. This includes both the reworking of catalogue products and the production of entirely new items.



Order: **G48**. d1 x d2 x L

Material:
Bronze + graphite inserts

Customised Guide Units

Upon request, Güvenal manufactures highly complex and precise guide elements based on customer specifications and provides measurement protocols if needed. This includes both the reworking of catalogue products and the production of entirely new items.

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Guiding Components

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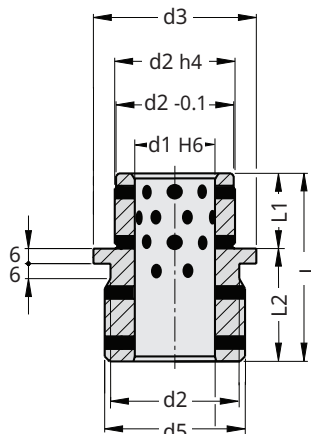
Working
Direction

Code: **G45**

Guide Bush with Centre Collar, Self-Lubricating

DIN 9834 / ISO 9448

Heat Resistance: < 150°C



Positioning :
Pls. do not make
direct fitting.
Precision assembly
recommended
with bronze / brass.



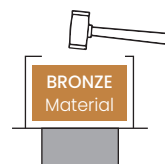
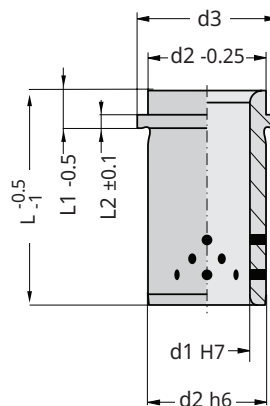
Working
Direction

Code: **G44**

Guide Bush with Collar, Self-Lubricating

DIN 9834 / ISO 9448

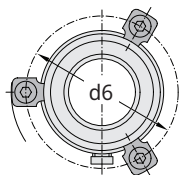
Heat Resistance: < 150°C



Positioning :
Pls. do not make
direct fitting.
Precision assembly
recommended
with bronze / brass.

d1	L	d2	d3	d5	d6	L1	L2
19	32	32	40	-	57.4	20	12
20	50			34			30
20	70						50
24	35	40	48	-	65.4	23	12
24	60			42			37
25	80						57
32	42	48	56	-	73.4	30	12
32	75			50			45
30	95						65
38	50	58	66	-	83.4	35	15
40	80			60			45
40	110						75
48	65	70	80	-	97.4	45	20
48	90			74			45
50	120						75

d1	L	d2	d3	L1	L2	d6
20	32	28	34	12	6.3	53.4
25	40	32	40	10		57.4
30	50	40	50	12		65.4
32	50	40	50	12		65.4
40	63	50	63	15		
50	71	63	71	17	10	122.4
63	80	80	90	19		142.4
80	100	100	112	22		167.4
100	125	125	140	21		202.4
125	160	160	180	30		242.4
160	200	200	220	32		



G101A
Clamping
Detail

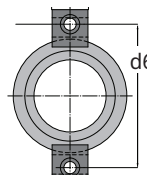
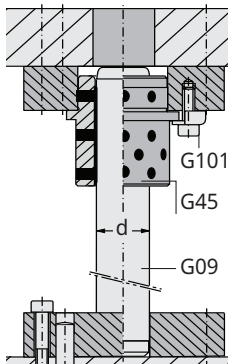


Order: **G45**. d1 x L



Material:

Bronze + Graphite inserts



G102
Clamping
Detail

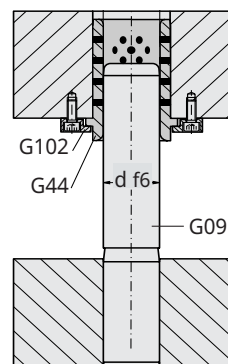


Order: **G44**. d1 x L



Material:

Bronze + Graphite inserts



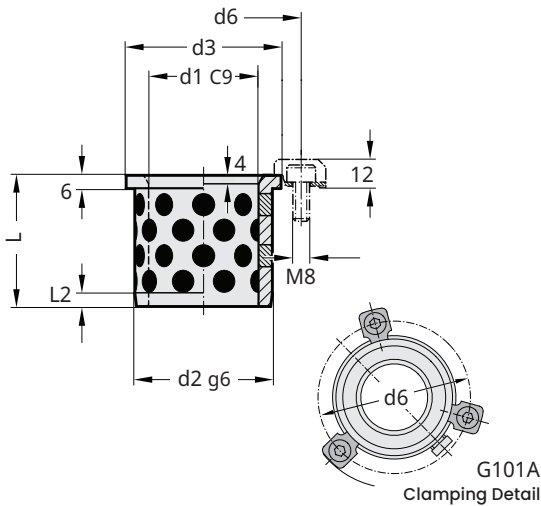


Guide Bush, Headed - NAAMS

Code: **G41**

Self-Lubricating

Heat Resistance: < 150°C



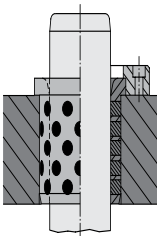
d1	L	d2	d3	L2	d6
25	40	32	40	4	57.4
32	50	40	50	4	67.4
40	55	50	63	5	80.4
50	63	63	71	6	88.4
63	75	80	90	8	107.4
80	90	100	112	10	129.4
100	115	125	140	12	157.4
125	138	160	180	12	197.4



Positioning :
Pls. do not make direct fitting.
Precision assembly recommended
with bronze / brass.

Order: **G41**. d1 x L

Material:
Bronze + Graphite inserts

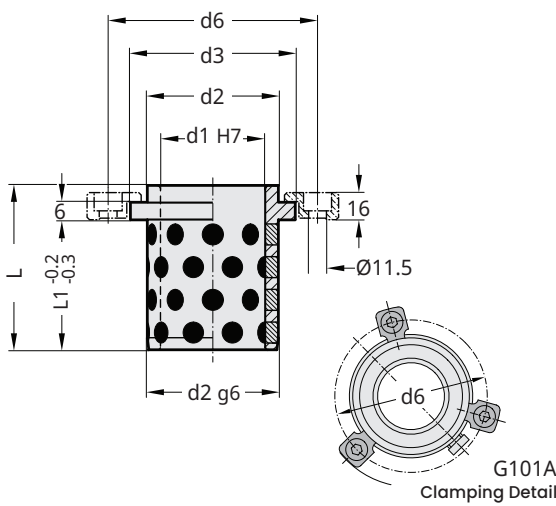


Guide Bush with Collar - NAAMS

Code: **G40**

Self-Lubricating

Heat Resistance: < 150°C



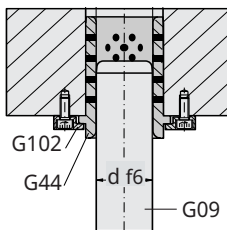
d1	L	d2	d3	d6	L1
20	40	28	34	51.4	30
25	40	32	40	57.4	30
30	50	40	50	67.4	40
32	50	40	50	67.4	40
40	63	50	63	80.4	50
50	71	63	71	88.4	56
63	80	80	90	107.4	63
80	100	100	112	129.4	80
100	125	125	140	157.4	106
125	160	160	180	197.4	132



Positioning :
Pls. do not make direct fitting.
Precision assembly recommended
with bronze / brass.

Order: **G40**. d1 x L

Material:
Bronze + Graphite inserts



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Guiding Components

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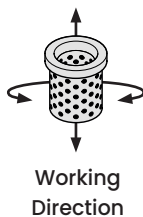
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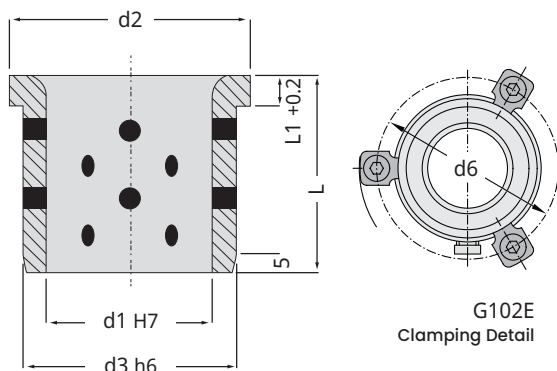


Guide Bush Headed

Code: **G42**

Self-Lubricating

Heat Resistance: < 150°C



d1	L	d3	d2	L1	d6
20	35	28	36	8	49.4
25	40	32	40		53.4
30	55	40	50		63.4
32	55	40	50		63.4
40	70	50	60		73.4
42	70	50	60		73.4
50	75	63	75	12	95.4
52	75	63	75		95.4
63	80	80	90		110.4
80	100	100	110		130.4
100	125	125	135		155.4



Order:
G42. d1 x L



Material:
Bronze + Graphite inserts



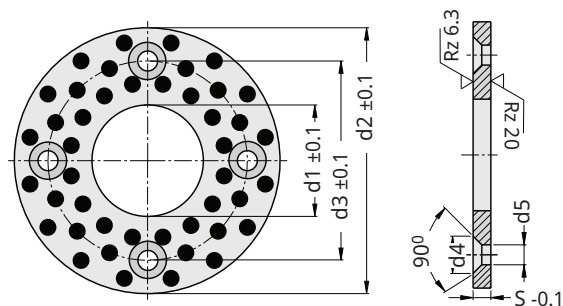
Thrust Washer - Perforated

Code: **G107**

Self-Lubricating

Heat Resistance: < 150°C

- * Spare part for in-die applications
- * Spare part in core systems etc. in injection moulds



d1	d2	d3	S	d4	d5	Hole
12.2	40	28	3	6.9	3	2
16.2	50	28	3	6.9	3	2
20.2	50	35	5	11.5	5.5	2
25.2	55	40	5	11.5	5.5	2
30.2	60	45	5	11.5	5.5	2
35.2	70	50	5	11.5	5.5	2
40.2	80	60	7	13.8	6.6	2
45.3	90	67.5	7	13.8	6.6	2
50.3	100	75	8	13.8	6.6	4
60.3	120	90	8	18.3	9	4
70.3	130	100	10	18.3	9	4
80.3	150	120	10	18.3	9	4



Order:
G107. d1 x d2



Material:
Bronze + Graphite inserts

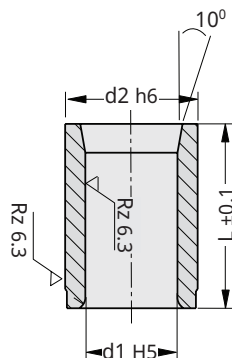
Guiding Components

NEW



Bronze Bush - Plain

Code: **G48 - 1**

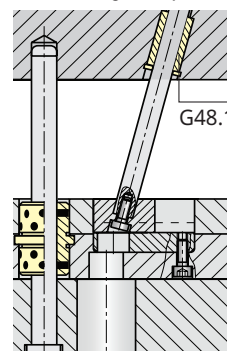


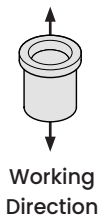
d1	L	d2
10	14	14
	20	
12	20	17
	25	
16	25	22
	32	



Order:
G48 - 1. d1 x L

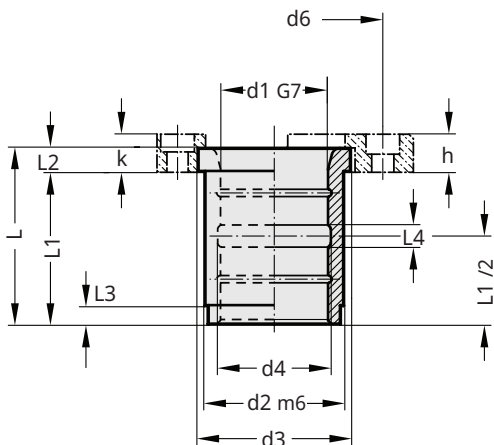
Mounting Example





Guide Bush, Bronze
Heat Resistance: < 150°C

Code: **G86**



d1	L	d2	d3	d4	d6	L1	L2	L3	L4	h	k
20	32	28	32	22	-	28	4	3	5	10	-
25	40	35	40	27	-	35	5	5	5	10	-
32	50	44	50	34	-	44	6	8	5	12	-
40	63	52	60	42	73.4	55	8	8	8	12	12
50	80	63	71	52	87.4	70	10	8	10	16	16
63	100	80	90	65	110.4	88	12	10	12	20	20
80	125	100	112	82	132.4	109	16	10	16	25	25
100	160	125	140	102	160.4	140	20	10	20	32	32



Positioning :
Pls. do not make direct fitting.
Precision assembly recommended
with bronze / brass.

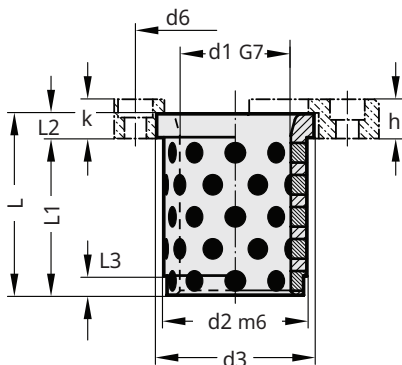


Order: **G86**. d1 x L

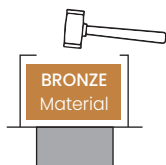


Guide Bush, Headed - CNOMO
Self-Lubricating

Code: **G43**



d1	L	d2	d3	d6	L1	L2	L3	h	k
20	32	28	32	-	28	4	3	10	-
25	40	35	40	-	35	5	5	10	-
32	50	44	50	-	44	6	5	12	-
40	63	52	60	73.4	55	8	8	12	12
50	80	63	71	87.4	70	10	8	16	16
63	100	80	90	110.4	88	12	10	20	20
80	125	100	112	132.4	109	16	10	25	25
100	160	125	140	160.4	140	20	10	32	32



Positioning :
Pls. do not make direct fitting.
Precision assembly recommended
with bronze / brass.



Order: **G43**. d1 x L



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